

27.1 The bias–variance trade-off

Chapter 26 showed where collinearity does its damage (Theorem 26.1.1). The least squares estimate remains unbiased, but along a direction $\mathbf{v}$ in which the regressors barely vary, the variance of $\mathbf{v}^\top \hat{\boldsymbol{\beta}}$ is σ^2 divided by a tiny eigenvalue; for a single coefficient the same damage appears as the factor $1/(1 - R_j^2)$ of Eq. 6.6.2, the variance inflation factor of Definition 26.2.1. No assumption of Parts II and III is violated. The Gauss–Markov theorem (Theorem 7.2.1) concerns the class of *linear unbiased* estimators, within which least squares cannot be improved, so improving on it means leaving the class. This section sets up the criterion that makes biased and unbiased estimators comparable.

27.1.1 The mean squared error matrix

Definition 27.1.1. Mean squared error

Let $\tilde{\boldsymbol{\theta}}$ be an estimator of $\boldsymbol{\theta} \in \mathbb{R}^q$ with finite second moments. Its **mean squared error matrix** is

$$\text{MSE}(\tilde{\boldsymbol{\theta}}) = \mathbb{E}[(\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta})(\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta})^\top],$$

its **bias** is $\boldsymbol{\delta} = \mathbb{E}(\tilde{\boldsymbol{\theta}}) - \boldsymbol{\theta}$, and its **total mean squared error** is $\text{tr MSE}(\tilde{\boldsymbol{\theta}}) = \mathbb{E}\|\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta}\|^2$. For estimators $\tilde{\boldsymbol{\theta}}_1, \tilde{\boldsymbol{\theta}}_2$ we say that $\tilde{\boldsymbol{\theta}}_1$ is **at least as good in the matrix sense** if $\text{MSE}(\tilde{\boldsymbol{\theta}}_2) - \text{MSE}(\tilde{\boldsymbol{\theta}}_1)$ is nonnegative definite.

All three quantities depend on the unknown parameter, so one estimator may be better for some parameter values and worse for others. For unbiased estimators the mean squared error matrix is the covariance matrix, and the matrix sense is the ordering in which Gauss–Markov declares least squares the winner (Theorem 7.2.1(d)).

Theorem 27.1.1. Bias–variance decomposition for linear estimators

- (a) For any estimator with finite second moments, $\text{MSE}(\tilde{\boldsymbol{\theta}}) = \text{Cov}(\tilde{\boldsymbol{\theta}}) + \boldsymbol{\delta}\boldsymbol{\delta}^\top$. For every $\mathbf{a} \in \mathbb{R}^q$, $\mathbb{E}(\mathbf{a}^\top \tilde{\boldsymbol{\theta}} - \mathbf{a}^\top \boldsymbol{\theta})^2 = \mathbf{a}^\top \text{MSE}(\tilde{\boldsymbol{\theta}}) \mathbf{a}$, and for every nonnegative definite $\mathbf{W}$, $\mathbb{E}[(\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta})^\top \mathbf{W}(\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta})] = \text{tr}(\mathbf{W} \text{MSE}(\tilde{\boldsymbol{\theta}}))$.
- (b) Suppose $\mathbb{E}(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ and $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$, and let $\tilde{\boldsymbol{\beta}} = \mathbf{A}\mathbf{Y}$ be a linear estimator of $\boldsymbol{\beta}$. Then $\boldsymbol{\delta} = (\mathbf{A}\mathbf{X} - \mathbf{I})\boldsymbol{\beta}$ and $\text{MSE}(\tilde{\boldsymbol{\beta}}) = \sigma^2 \mathbf{A}\mathbf{A}^\top + (\mathbf{A}\mathbf{X} - \mathbf{I})\boldsymbol{\beta}\boldsymbol{\beta}^\top(\mathbf{A}\mathbf{X} - \mathbf{I})^\top$.
- (c) In the setting of (b), let $\mathbf{X}$ have full column rank, let $\hat{\boldsymbol{\beta}}$ be least squares, and put $\boldsymbol{\Delta}_\mathbf{A} = \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1} - \sigma^2 \mathbf{A}\mathbf{A}^\top$, the covariance that $\tilde{\boldsymbol{\beta}}$ saves. The following are equivalent:
 - (i) $\tilde{\boldsymbol{\beta}}$ is at least as good as $\hat{\boldsymbol{\beta}}$ in the matrix sense;
 - (ii) $\mathbb{E}(\mathbf{a}^\top \tilde{\boldsymbol{\beta}} - \mathbf{a}^\top \boldsymbol{\beta})^2 \leq \text{Var}(\mathbf{a}^\top \hat{\boldsymbol{\beta}})$ for every $\mathbf{a}$;
 - (iii) $\mathbb{E}[(\tilde{\boldsymbol{\beta}} - \boldsymbol{\beta})^\top \mathbf{W}(\tilde{\boldsymbol{\beta}} - \boldsymbol{\beta})] \leq \mathbb{E}[(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})^\top \mathbf{W}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})]$ for every nonnegative definite $\mathbf{W}$;
 - (iv) $\boldsymbol{\Delta}_\mathbf{A}$ is nonnegative definite, $\boldsymbol{\delta} \in \mathcal{C}(\boldsymbol{\Delta}_\mathbf{A})$ and $\boldsymbol{\delta}^\top \boldsymbol{\Delta}_\mathbf{A}^+ \boldsymbol{\delta} \leq 1$.
When $\boldsymbol{\Delta}_\mathbf{A}$ is positive definite, (iv) reads $\boldsymbol{\delta}^\top \boldsymbol{\Delta}_\mathbf{A}^{-1} \boldsymbol{\delta} \leq 1$.

Proof.

- (a) Write $\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta} = (\tilde{\boldsymbol{\theta}} - \mathbb{E}\tilde{\boldsymbol{\theta}}) + \boldsymbol{\delta}$. The cross terms have expectation zero, which gives the decomposition. The scalar identities follow from $\mathbb{E}(\mathbf{a}^\top \mathbf{w})^2 = \mathbf{a}^\top \mathbb{E}(\mathbf{w}\mathbf{w}^\top) \mathbf{a}$ and

$$\mathbf{w}^\top \mathbf{W} \mathbf{w} = \text{tr}(\mathbf{W} \mathbf{w} \mathbf{w}^\top) \text{ with } \mathbf{w} = \tilde{\boldsymbol{\theta}} - \boldsymbol{\theta}.$$

- (b) $E(\mathbf{A}\mathbf{Y}) = \mathbf{A}\mathbf{X}\boldsymbol{\beta}$ and $\text{Cov}(\mathbf{A}\mathbf{Y}) = \sigma^2 \mathbf{A}\mathbf{A}^\top$ (Theorem 2.2.1); apply (a).
- (c) By (a), $\text{MSE}(\hat{\boldsymbol{\beta}}) - \text{MSE}(\tilde{\boldsymbol{\beta}}) = \boldsymbol{\Delta}_\mathbf{A} - \boldsymbol{\delta}\boldsymbol{\delta}^\top$, and (i) says this matrix is nonnegative definite. By (a), (ii) says $\mathbf{a}^\top(\boldsymbol{\Delta}_\mathbf{A} - \boldsymbol{\delta}\boldsymbol{\delta}^\top)\mathbf{a} \geq 0$ for all $\mathbf{a}$, which is (i). Next, (i) implies (iii) because $\text{tr}(\mathbf{W}\mathbf{N}) \geq 0$ when $\mathbf{W}$ and $\mathbf{N}$ are both nonnegative definite: with $\mathbf{W} = \sum_k \omega_k \mathbf{q}_k \mathbf{q}_k^\top$ (Theorem 1.7.2), $\text{tr}(\mathbf{W}\mathbf{N}) = \sum_k \omega_k \mathbf{q}_k^\top \mathbf{N} \mathbf{q}_k \geq 0$. And (iii) with $\mathbf{W} = \mathbf{a}\mathbf{a}^\top$ is (ii). For (i) $\Leftrightarrow$ (iv), put $\mathbf{N} = \boldsymbol{\Delta}_\mathbf{A} - \boldsymbol{\delta}\boldsymbol{\delta}^\top$. If $\mathbf{N}$ is nonnegative definite, so is $\boldsymbol{\Delta}_\mathbf{A} = \mathbf{N} + \boldsymbol{\delta}\boldsymbol{\delta}^\top$. If $\boldsymbol{\delta} \notin \mathcal{C}(\boldsymbol{\Delta}_\mathbf{A})$, there is $\mathbf{w} \in \mathcal{N}(\boldsymbol{\Delta}_\mathbf{A})$ with $\mathbf{w}^\top \boldsymbol{\delta} \neq 0$ (Proposition 1.5.1), and then $\mathbf{w}^\top \mathbf{N} \mathbf{w} = -(\mathbf{w}^\top \boldsymbol{\delta})^2 < 0$. So (i) forces $\boldsymbol{\delta} = \boldsymbol{\Delta}_\mathbf{A} \mathbf{c}$ for some $\mathbf{c}$, and $\mathbf{c}^\top \mathbf{N} \mathbf{c} = \mathbf{c}^\top \boldsymbol{\Delta}_\mathbf{A} \mathbf{c} (1 - \mathbf{c}^\top \boldsymbol{\Delta}_\mathbf{A} \mathbf{c}) \geq 0$ gives $\boldsymbol{\delta}^\top \boldsymbol{\Delta}_\mathbf{A}^+ \boldsymbol{\delta} = \mathbf{c}^\top \boldsymbol{\Delta}_\mathbf{A} \mathbf{c} \leq 1$ (when $\mathbf{c}^\top \boldsymbol{\Delta}_\mathbf{A} \mathbf{c} = 0$ there is nothing to prove). Here $\boldsymbol{\delta}^\top \boldsymbol{\Delta}_\mathbf{A}^+ \boldsymbol{\delta} = \mathbf{c}^\top \boldsymbol{\Delta}_\mathbf{A} \boldsymbol{\Delta}_\mathbf{A}^+ \boldsymbol{\Delta}_\mathbf{A} \mathbf{c} = \mathbf{c}^\top \boldsymbol{\Delta}_\mathbf{A} \mathbf{c}$ by the first Penrose condition. Conversely, under (iv) the Cauchy–Schwarz inequality for the semi-inner product $(\mathbf{w}, \mathbf{z}) \mapsto \mathbf{w}^\top \boldsymbol{\Delta}_\mathbf{A} \mathbf{z}$ gives, for every $\mathbf{w}$,

$$(\mathbf{w}^\top \boldsymbol{\delta})^2 = (\mathbf{w}^\top \boldsymbol{\Delta}_\mathbf{A} \mathbf{c})^2 \leq (\mathbf{w}^\top \boldsymbol{\Delta}_\mathbf{A} \mathbf{w})(\mathbf{c}^\top \boldsymbol{\Delta}_\mathbf{A} \mathbf{c}) \leq \mathbf{w}^\top \boldsymbol{\Delta}_\mathbf{A} \mathbf{w},$$

so $\mathbf{w}^\top \mathbf{N} \mathbf{w} \geq 0$. When $\boldsymbol{\Delta}_\mathbf{A}$ is invertible, $\boldsymbol{\Delta}_\mathbf{A}^+ = \boldsymbol{\Delta}_\mathbf{A}^{-1}$; this case is also Lemma 19.1.4. ■

Part (c) makes the trade-off precise: a biased linear estimator is better in every direction and under every quadratic loss exactly when its squared bias, measured in the metric of the covariance it saves, is at most one. Since $\boldsymbol{\delta} = (\mathbf{A}\mathbf{X} - \mathbf{I})\boldsymbol{\beta}$, condition (iv) holds when $\boldsymbol{\beta}$ is close to what the estimator shrinks towards, and fails far from it. No linear estimator with $\mathbf{A}\mathbf{X} \neq \mathbf{I}$ wins everywhere (Exercise (B1)). Theorem 19.1.5 was the first instance: the short regression is a biased linear estimator, and its winning region $\gamma \leq 1$ is condition (iv) for that choice of $\mathbf{A}$.

27.1.2 What unbiasedness buys, and what it costs

Unbiasedness concerns an estimator's centre, not its accuracy, and it does not survive transformation: $\|\hat{\boldsymbol{\beta}}\|^2$ overestimates $\|\boldsymbol{\beta}\|^2$ by $\sigma^2 \text{tr}(\mathbf{X}^\top \mathbf{X})^{-1}$ on average (Exercise (B3)), which under collinearity is enormous. Least squares estimates are systematically *too long*, and shrinking them is a correction, not a distortion. When many parameters are estimated at once, small biases spread over many coordinates can buy a large cut in total variance; Section 27.5 turns this into a theorem. The cost is in inference: the bias depends on the unknown $\boldsymbol{\beta}$, so an interval “estimate $\pm$ multiple of standard error” centred on a shrinkage estimate has no guaranteed coverage. Chapter 28 and Chapter 29 return to inference after shrinkage and selection.

27.1.3 Shrinkage in canonical coordinates

The collinearity problem has a natural coordinate system. Let $\mathbf{X}$ have full column rank and let $\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}^\top$ be its thin singular value decomposition (Theorem 1.8.1), with $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_p]$ and $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_p]$ having orthonormal columns and $\mathbf{D} = \text{diag}(d_1, \dots, d_p)$, $d_1 \geq \dots \geq d_p > 0$. We write d_j instead of the σ_j of Chapter 1 because σ is the error standard deviation. Put

$$\boldsymbol{\alpha} = \mathbf{V}^\top \boldsymbol{\beta}, \quad \hat{\boldsymbol{\alpha}} = \mathbf{V}^\top \hat{\boldsymbol{\beta}} = \mathbf{D}^{-1} \mathbf{U}^\top \mathbf{Y}, \quad c_j = \mathbf{u}_j^\top \mathbf{Y} = d_j \hat{\alpha}_j. \quad (27.1.1)$$

The α_j are the **canonical coefficients**: the coordinates of $\boldsymbol{\beta}$ along the eigenvectors $\mathbf{v}_j$ of $\mathbf{X}^\top \mathbf{X}$, whose eigenvalues are d_j^2 . By Theorem 10.4.1(d), the $\hat{\alpha}_j$ are uncorrelated with $E(\hat{\alpha}_j) = \alpha_j$ and $\text{Var}(\hat{\alpha}_j) = \sigma^2/d_j^2$. A small d_j is a direction $\mathbf{v}_j$ along which the regressors hardly move, and along

which the data therefore say little (Proposition 19.6.1).

Many shrinkage estimators act on the canonical coordinates one at a time. For numbers $f_1, \dots, f_p$, called **shrinkage factors** or filter factors, define

$$\tilde{\boldsymbol{\beta}}_{\mathbf{f}} = \sum_{j=1}^p f_j \hat{\alpha}_j \mathbf{v}_j = \mathbf{V} \mathbf{F} \hat{\boldsymbol{\alpha}}, \quad \mathbf{F} = \text{diag}(f_1, \dots, f_p).$$

Least squares has every $f_j = 1$. Principal component regression (Section 27.3) has $f_j \in \{0, 1\}$, and ridge regression (Section 27.2) has $f_j = d_j^2 / (d_j^2 + \lambda)$. The signal-to-noise ratio of direction j is

$$\tau_j^2 = \frac{\alpha_j^2}{\text{Var}(\hat{\alpha}_j)} = \frac{d_j^2 \alpha_j^2}{\sigma^2},$$

the squared mean of the standardized coordinate $\hat{\alpha}_j / \sqrt{\text{Var}(\hat{\alpha}_j)}$. Under normal errors it is the noncentrality of $c_j^2 / \sigma^2 \sim \chi^2(1, \tau_j^2)$.

Proposition 27.1.2. Shrinking along the principal directions

Let $\mathbf{E}(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$, $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ and $\text{rank}(\mathbf{X}) = p$. For fixed factors f_j :

- (a) $\mathbf{E}\|\tilde{\boldsymbol{\beta}}_{\mathbf{f}} - \boldsymbol{\beta}\|^2 = \sum_j \{f_j^2 \sigma^2 / d_j^2 + (1 - f_j)^2 \alpha_j^2\}$;
- (b) $\mathbf{E}\|\mathbf{X}\tilde{\boldsymbol{\beta}}_{\mathbf{f}} - \mathbf{X}\boldsymbol{\beta}\|^2 = \sum_j \{f_j^2 \sigma^2 + (1 - f_j)^2 d_j^2 \alpha_j^2\}$;
- (c) both are minimized by the same factors $f_j^* = \tau_j^2 / (1 + \tau_j^2)$, and the minima are $\sum_j (\sigma^2 / d_j^2) \tau_j^2 / (1 + \tau_j^2)$ and $\sigma^2 \sum_j \tau_j^2 / (1 + \tau_j^2)$;
- (d) replacing $f_j = 1$ by some $f_j \in [0, 1]$, with the other factors fixed, lowers both quantities iff $\tau_j^2 < (1 + f_j) / (1 - f_j)$. In particular dropping direction j ($f_j = 0$) helps iff $\tau_j^2 < 1$.

Proof. $\tilde{\boldsymbol{\beta}}_{\mathbf{f}} - \boldsymbol{\beta} = \mathbf{V}(\mathbf{F}\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha})$ and $\mathbf{V}$ is orthogonal, so $\|\tilde{\boldsymbol{\beta}}_{\mathbf{f}} - \boldsymbol{\beta}\|^2 = \sum_j (f_j \hat{\alpha}_j - \alpha_j)^2$. Each term has expectation $\text{Var}(f_j \hat{\alpha}_j) + (f_j \alpha_j - \alpha_j)^2 = f_j^2 \sigma^2 / d_j^2 + (1 - f_j)^2 \alpha_j^2$, which is (a). For (b), $\mathbf{X}(\tilde{\boldsymbol{\beta}}_{\mathbf{f}} - \boldsymbol{\beta}) = \mathbf{U} \mathbf{D}(\mathbf{F}\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha})$ and $\mathbf{U}$ has orthonormal columns, so each term of (a) is multiplied by d_j^2 . (c) Each term of (a) is a convex quadratic in f_j with derivative $2f_j \sigma^2 / d_j^2 - 2(1 - f_j) \alpha_j^2$, which vanishes at $f_j = \alpha_j^2 / (\alpha_j^2 + \sigma^2 / d_j^2) = \tau_j^2 / (1 + \tau_j^2)$. Terms of (b) are those of (a) times d_j^2 , so they have the same minimizer; substituting gives the minima. (d) The j th term of (a) falls below its value σ^2 / d_j^2 at $f_j = 1$ iff $(1 - f_j)^2 \alpha_j^2 < (1 - f_j^2) \sigma^2 / d_j^2$. Dividing by $(1 - f_j) \sigma^2 / d_j^2 > 0$ gives $(1 - f_j) \tau_j^2 < 1 + f_j$. The same holds for (b). ■

The ideal shrinkage depends on τ_j^2 , which combines the spread of the regressors in direction j and the size of the coefficient there, so a direction with tiny d_j but large α_j should be kept. The ideal factors are the same for estimating $\boldsymbol{\beta}$ and $\mathbf{X}\boldsymbol{\beta}$. The oracle factors depend on $\boldsymbol{\beta}$ and σ^2 ; each method of this chapter chooses factors from the data instead, ridge by one tuning constant, principal components by a cut-off, and James–Stein by an unbiased estimate of the risk. Part (d) with $f_j = 0$ is the criterion of Proposition 10.4.3.

Example. Two nearly collinear regressors

Take $n = 50$ observations of two centred regressors, scaled so that $\mathbf{X}^\top \mathbf{X} = n \mathbf{R}$ where $\mathbf{R}$ is a

correlation matrix with off-diagonal entry 0.95. Let $\sigma = 1$ and $\boldsymbol{\beta} = (1, 0.5)^\top$. The eigenvalues of $\mathbf{X}^\top \mathbf{X}$ are $d_1^2 = n(1+0.95) = 97.5$ and $d_2^2 = n(1-0.95) = 2.5$, with eigenvectors $(1, 1)^\top/\sqrt{2}$ and $(1, -1)^\top/\sqrt{2}$. The canonical coefficients have sizes $|\alpha_1| = 1.0607$ and $|\alpha_2| = 0.3536$, and the signal-to-noise ratios are $\tau_1^2 = 109.7$ and $\tau_2^2 = 0.3125$. The first direction, the sum of the two regressors, is estimated precisely. The second, their difference, is not.

Estimator	Factors (f_1, f_2)	Total MSE
least squares	(1, 1)	0.4103
drop the second direction	(1, 0)	0.1353
halve the second direction	(1, 0.5)	0.1415
oracle f_j^*	(0.991, 0.238)	0.1054

Dropping the difference direction cuts the total mean squared error by two thirds, because $\tau_2^2 < 1$, and wins in the matrix sense too. Here $\Delta_{\mathbf{A}} = \sigma^2 \mathbf{v}_2 \mathbf{v}_2^\top / d_2^2$ is singular, the bias is $\boldsymbol{\delta} = -\alpha_2 \mathbf{v}_2 \in \mathcal{C}(\Delta_{\mathbf{A}})$, and $\boldsymbol{\delta}^\top \Delta_{\mathbf{A}}^+ \boldsymbol{\delta} = \alpha_2^2 d_2^2 / \sigma^2 = \tau_2^2 = 0.3125 \leq 1$. So by [Theorem 27.1.1\(c\)](#), every linear combination $\mathbf{a}^\top \boldsymbol{\beta}$ is estimated at least as accurately. Had the truth been $\boldsymbol{\beta} = (1, -0.5)^\top$, with the same length but pointing along the difference, then $\tau_2^2 = 2.5 \times 1.125 = 2.81$ and dropping would have increased the total error.

```
import numpy as np
n, rho, sigma = 50, 0.95, 1.0
XtX = n * np.array([[1.0, rho], [rho, 1.0]])
beta = np.array([1.0, 0.5])

evals, V = np.linalg.eigh(XtX)
order = np.argsort(evals)[::-1]
d2, V = evals[order], V[:, order]          # d_1^2 >= d_2^2, eigenvectors v_1, v_2
alpha = V.T @ beta                        # canonical coefficients
tau2 = d2 * alpha ** 2 / sigma ** 2        # squared signal-to-noise ratio of each
direction

def mse(f):
    """Total MSE of the estimator that multiplies alpha_hat_j by f_j."""
    f = np.asarray(f, dtype=float)
    return np.sum(f ** 2 * sigma ** 2 / d2 + (1 - f) ** 2 * alpha ** 2)

f_oracle = tau2 / (1 + tau2)
for label, f in [("least squares", [1, 1]), ("drop v_2", [1, 0]),
                 ("halve v_2", [1, 0.5]), ("oracle", f_oracle)]:
    print(f"{label:14s} f = {np.round(f, 3)}    MSE = {mse(f):.4f}")
print("tau^2 =", np.round(tau2, 3))
```

[Run this code online](#)

27.1.4 Exercises

27.1.4.1 A. Check your understanding

Exercise. A1

Let $\bar{Y}$ be the mean of n independent observations with mean μ and variance σ^2 . For a

constant c , find the mean squared error of $c\bar{Y}$ as an estimator of μ . Show that it is minimized at $c^* = \mu^2/(\mu^2 + \sigma^2/n)$ and that $c\bar{Y}$ beats $\bar{Y}$ iff $(1 - c)\mu^2 < (1 + c)\sigma^2/n$. Why is c^* not an estimator?

Exercise. A2

In [Example \(Two nearly collinear regressors\)](#), verify the least squares total mean squared error $\sigma^2(1/d_1^2 + 1/d_2^2) = 0.4103$ by hand, and check the entry for “drop the second direction” using [Proposition 27.1.2\(a\)](#).

27.1.4.2 B. Practice

Exercise. B1

Let $\mathbf{X}$ have full column rank and let $\tilde{\boldsymbol{\beta}} = \mathbf{A}\mathbf{Y}$ with $\mathbf{A}\mathbf{X} \neq \mathbf{I}$. Show that there is a $\boldsymbol{\beta}$ for which $E\|\tilde{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 > E\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2$. So no biased linear estimator dominates least squares.

Solution. By [Theorem 27.1.1\(b\)](#), $E\|\tilde{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 = \sigma^2 \text{tr}(\mathbf{A}\mathbf{A}^\top) + \|(\mathbf{A}\mathbf{X} - \mathbf{I})\boldsymbol{\beta}\|^2$. The first term does not depend on $\boldsymbol{\beta}$. Since $\mathbf{A}\mathbf{X} - \mathbf{I} \neq \mathbf{0}$ there is a unit vector $\mathbf{v}$ with $(\mathbf{A}\mathbf{X} - \mathbf{I})\mathbf{v} \neq \mathbf{0}$. Along $\boldsymbol{\beta} = t\mathbf{v}$ the second term is $t^2\|(\mathbf{A}\mathbf{X} - \mathbf{I})\mathbf{v}\|^2$, which exceeds the constant $\sigma^2 \text{tr}(\mathbf{X}^\top \mathbf{X})^{-1}$ for large t .

Exercise. B2

A **linear smoother** estimates the mean vector $\boldsymbol{\mu} = E(\mathbf{Y})$ by $\mathbf{S}\mathbf{Y}$ for a fixed $n \times n$ matrix $\mathbf{S}$. Show that $E\|\mathbf{S}\mathbf{Y} - \boldsymbol{\mu}\|^2 = \sigma^2 \text{tr}(\mathbf{S}\mathbf{S}^\top) + \|(\mathbf{I} - \mathbf{S})\boldsymbol{\mu}\|^2$ when $\text{Cov}(\mathbf{Y}) = \sigma^2\mathbf{I}$. For the projection $\mathbf{S} = \mathbf{M}$ and $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$, recover $\sigma^2 r$ with $r = \text{rank}(\mathbf{X})$.

Solution. $\mathbf{S}\mathbf{Y} - \boldsymbol{\mu} = \mathbf{S}(\mathbf{Y} - \boldsymbol{\mu}) - (\mathbf{I} - \mathbf{S})\boldsymbol{\mu}$. The first term has mean zero and covariance $\sigma^2\mathbf{S}\mathbf{S}^\top$, so its expected squared length is $\sigma^2 \text{tr}(\mathbf{S}\mathbf{S}^\top)$ ([Theorem 2.3.1](#)). The second term is constant, and the cross term has mean zero. For $\mathbf{S} = \mathbf{M}$, $\text{tr}(\mathbf{M}\mathbf{M}^\top) = \text{tr} \mathbf{M} = r$ and $(\mathbf{I} - \mathbf{M})\boldsymbol{\mu} = \mathbf{0}$.

Exercise. B3

Show that $E\|\hat{\boldsymbol{\beta}}\|^2 = \|\boldsymbol{\beta}\|^2 + \sigma^2 \sum_j d_j^{-2}$. For which c does $c\hat{\boldsymbol{\beta}}$ have smaller total mean squared error than $\hat{\boldsymbol{\beta}}$? Express the answer through $\|\boldsymbol{\beta}\|^2$ and $\sigma^2 \text{tr}(\mathbf{X}^\top \mathbf{X})^{-1}$.

27.1.4.3 C. Going deeper

Exercise. C1

Least squares is equivariant under rescaling: replacing $\mathbf{X}$ by $\mathbf{X}\mathbf{G}^{-1}$, with $\mathbf{G}$ diagonal and positive, turns the estimate into $\mathbf{G}\hat{\boldsymbol{\beta}}$. By rescaling one column in [Example \(Two nearly](#)

collinear regressors) by 10, show that $\tilde{\boldsymbol{\beta}}_{\mathbf{f}}$ with $f_1 = 1$, $f_2 = 0$ is not equivariant. Why are shrinkage estimators usually applied to standardized regressors?